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Deep Fibration Attention

-meta abduction semantics for query updating

Gary Nan Tie, Aug 31, 2025

Abstract

In fibration attention, a lifting simultaneously updates a query and abductively finds a key to explain a value. This all depends on the key-value pairing (k, v) . By determining other keys that are highly similar to k , to also explain v , we can enhance key-value semantics to achieve more meaningful attention. Moreover, we can recursively layer keys upon keys in a tree, to deepen the set of value explanations.

In latent space $\mathbb{R}^d$, let subsets Q , K and V denote queries, keys and values respectively, and let Q' be a copy of Q , denoting updated queries.

Data: $q_1, \dots, q_n \in Q$

$k_1, \dots, k_m \in K$

$v_1, \dots, v_m \in V$

Let $L: \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^+$ be a loss function.

For $\varepsilon > 0$, define $x =_\varepsilon y$ iff $L(x, y) < \varepsilon$.

Suppose there exists $\varepsilon > 0$ such that we can learn a neural network $p: K \rightarrow V$ with $p(k_j) =_\varepsilon v_j$, $j \in [m]$, a key-value semantic pairing.

Transformer attention via fibration:

Template diagram of neural networks, commuting up to $\approx_\varepsilon$,

that updates and refines a query in an attention layer:

$$\begin{array}{ccc}
 & m \otimes 1 & \\
 Q \otimes K & \xrightarrow{\quad} & Q' \otimes K \\
 1 \otimes p \downarrow \sigma & \dashrightarrow & \downarrow 1 \otimes p \\
 Q \otimes V & \xrightarrow{m \otimes 1} & Q' \otimes V
 \end{array} \quad (+)$$

$$\text{Learn } \sigma: Q \otimes V \rightarrow Q' \otimes K$$

so that (+) commutes up to $\approx_\varepsilon$, that is

$$\text{i) } \sigma \begin{bmatrix} q_i \\ p(k_j) \end{bmatrix} \approx_\varepsilon \begin{bmatrix} q'_i \\ k_j \end{bmatrix} \quad \text{where } q'_i = m(q_i)$$

$$\text{ii) } \pi_1 \sigma \begin{bmatrix} q_i \\ v_j \end{bmatrix} \approx_\varepsilon q'_i$$

$$\text{iii) } p \pi_2 \sigma \begin{bmatrix} q_i \\ v_j \end{bmatrix} \approx_\varepsilon v_j$$

Query update step from q_i to $\tilde{q}_i$:

Learn neural network $\sigma: Q \otimes V \rightarrow Q' \otimes K$ from (†)

$$\text{with } m(q_i) = q'_i \triangleq \sum_{j=1}^m \alpha_{ij} v_j,$$

where $\alpha_{ij} \triangleq \text{softmax}_j \text{score}(q_i, k_j)$,

score a similarity function, like a reproducing kernel.

Let $j^* = \arg \max_j \alpha_{ij}$

Define update of q_i to be $\tilde{q}_i = \pi_1 \sigma \begin{bmatrix} q'_i \\ v_{j^*} \end{bmatrix}$.

To refine query q_i , replace q_i with $\tilde{q}_i$, and successively update, until perplexity is acceptable.

Note that instead of applying a feed forward neural network to q'_i , as in generic transformer attention, we learn a ~~lift~~ lifting neural network σ , for fibration $1 \otimes p$.

Extending key-value semantics:

We want to extend the semantics of our key-value pairing $p: K \rightarrow V$ ~~into~~ which satisfies $p(k_j) =_\varepsilon v_j$.

Now let score be a similarity function between keys.

Fix $j \in [m]$, define probabilities

$$\alpha_{jl} = \text{softmax}_l \text{ score}(k_j, k_l).$$

For $0 < \lambda < 1$, let $B_{\lambda,j} = \{k_l : \alpha_{jl} > \lambda\}$

Learn neural network $P_\lambda: K \rightarrow V$

such that $P_\lambda(B_{\lambda,j}) =_\varepsilon v_j$, $j \in [m]$.

p explains value v_j with key k_j , while

P_λ explains value v_j with keys highly similar to k_j ,

an enhanced semantics, so to speak.

Now for each $k_l \in B_{\lambda,j}$ we can similarly define $B_{\lambda,l}$

and define $B_j^1 = \bigcup B_{\lambda,l}$.

Note that $k_j \in B_{\lambda,j} \subset B_j^1$.

Learn $p^1: K \rightarrow V$ such that $p^1(B_j^1) =_E v_j, j \in [m]$.

Now for each $k_t \in B_j^1$, define $B_{\lambda,t}$ and let $B_j^2 = \bigcup B_{\lambda,t}$.

Learn $p^2: K \rightarrow V$ such that $p^2(B_j^2) =_E v_j, j \in [m]$,

and so on, ever enlarging the set of keys related to k_j .

Summary: A key-value pairing $p: K \rightarrow V$ giving abductive semantics to transformer attention can be extended to

$p_\lambda: K \rightarrow V, 0 < \lambda < 1$, which broadens the set of keys

available to explain a value. By replacing p with p_λ

in (†), we can learn enhanced update lifting σ_λ .

Moreover, we can recursively iterate this key enlargement

process to successively learn semantics $p^1, p^2, \dots$

References

[1] 'Contextual Fibration Attention'

Gary Nan Tie, Aug 29, 2025

DOI: 10.13140/RG.2.2.12205.35044

In a query-key-value attention model, the key-value pairing conveys language semantics. Query context also carries meaning.

We introduce an abductive fibration attention model that holistically captures both. A fibration lifting, jointly learns from query context and language semantics (that abductively finds a key to best explain the returned value).

[2] 'Multi-head abductive attention'

Gary Nan Tie, Aug 25, 2025

DOI: 10.13140/RG.2.2.20597.23521

Transformer multi-head attention can focus on different parts of an input query sequence, however without regard to potentially different key-value semantics. This new primitive for formal query updating, constructs updates to a query in parallel, by learning multi-head abductive fibration liftings, to refine similarity based attention, utilizing the semantics of multiple key-value pairings.

[3] 'Formal query updating to generate problem specific fibration abduction attention algorithms'

Gary Nan Tie, Aug 23, 2025

DOI: 10.13140/RG.2.2.33029.41446

A formalism to construct problem specific query update algorithms founded on standard attention and fibration attention primitives.

[4] 'FibChat - It makes stuff up, but with structure.'

Gary Nan Tie · July 2025

DOI: 10.13140/RG.2.2.10549.59364/2

FibChat is a LLM based on fibration attention, that enables coherent structured mitigation of hallucinations.

[5] 'Fibrations explain all you need!

- Fibrations, Abduction, Attention

Gary Nan Tie, Mar 4, 2025

DOI: 10.13140/RG.2.2.35984.52488

To conceptualize and perform abductive machine learning, categorical fibrations are proposed, as they coherently give both context and structure for abductive reasoning via an attention mechanism. Moreover, 2-categorical lifting compatibility conditions ensure consistent hierarchical and parallel explanations.